Shaheed Zulfikar Ali Bhutto Institute of Science & Technology
COMPUTER SCIENCE DEPARTMENT | IDS BSSE-8 | SZABIST-ISB

Introduction to Data Science

Assignment #04 — Continuation of Assignment #03

AQI-Based Adaptive Traffic-Control System Using Reinforcement Learning

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Registration Number:	2280219
GitHub Repository:	https://github.com/mohammadqbbaig/AQI-Data-Science-Project.git
Dataset Used:	Global Urban AQI Dataset (2015-2025)
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1. Connection with Assignment 3

Assignment 3 involved a complete introductory data science workflow on the **Global Urban Air Quality Index Dataset (2015–2025)**, covering data cleaning, exploratory visualization, AQI category creation, K-Nearest Neighbours (KNN), Naïve Bayes, K-Means clustering, and Principal Component Analysis (PCA). Assignment 4 extends that same project by adding a Reinforcement Learning (RL) component for adaptive traffic control.

Assets Reused from Assignment 3:

Asset	How It Is Used in Assignment 4
Cleaned dataset (global_urban_aqi_dataset.csv)	Environment input for RL training & testing
AQI numerical values	Basis for defining RL states (Low / Medium / High AQI)
AQI category column	Directly mapped to RL state integers 0, 1, 2
GitHub repository	Assignment 4 notebook added as a new file in the same repo
Visualizations (charts folder)	RL charts saved alongside Assignment 3 charts

2. RL Problem Statement

This project extends the AQI data science analysis by designing a simplified reinforcement learning system for adaptive traffic control. The RL agent observes real AQI readings from the global urban dataset and learns whether a city should apply **No Restriction**, **Partial Restriction**, or a **High-Pollution Alert**. The objective is to understand how an agent can learn a sound decision policy through trial-and-error, using rewards that balance public-health protection with traffic-disruption cost.

3. Reinforcement Learning Component Definitions

RL Component	Definition in This Assignment
Agent	The adaptive traffic-control decision maker.
Environment	AQI conditions drawn from the cleaned global AQI dataset.
State	AQI level: Low AQI (≤100), Medium AQI (101–200), High AQI (>200).
Action	No Restriction (0), Partial Restriction (1), High-Pollution Alert (2).
Reward	Numerical score encouraging suitable action and penalising unsuitable action.
Policy	The learned mapping from each AQI state to the best traffic action.
Episode	One full training cycle iterating over all AQI records in the dataset.

4. AQI State Creation

A 3-state design (recommended for an introductory course) was adopted. Each AQI reading is mapped to one of three discrete states:

AQI Condition	AQI Range	RL State	Integer ID
Good / Moderate	0 – 100	Low AQI	0
Unhealthy for Sensitive / Unhealthy	101 – 200	Medium AQI	1
Very Unhealthy / Hazardous	201 +	High AQI	2

5. Action Space

Action ID	Action Name	Meaning
0	No Restriction	Normal traffic flow; no public warning required.
1	Partial Restriction	Moderate control — reduce heavy traffic, advise sensitive groups, limit high-emission vehicles.
2	High-Pollution Alert	Strong warning or restriction during dangerous AQI conditions.

6. Reward System

The reward table below encodes domain knowledge: over-restricting traffic during clean air earns a penalty, while taking no action during hazardous pollution earns the strongest penalty.

AQI State	No Restriction	Partial Restriction	High-Pollution Alert	Best Action
Low AQI	+10	–2	–5	No Restriction
Medium AQI	–6	+10	+2	Partial Restriction
High AQI	–10	–3	+10	High-Pollution Alert

7. Q-Learning Implementation Summary

A simple tabular Q-learning algorithm was implemented using NumPy, Pandas, and Matplotlib only — no advanced RL libraries were required.

Hyperparameter	Value	Reason
Learning rate (α)	0.1	Slow, stable updates — prevents overshooting.
Discount factor (γ)	0.9	Values future rewards; suitable for a static-state problem.
Exploration rate (ε)	0.1	10 % random exploration; 90 % exploitation of best known action.
Episodes	500	Sufficient for convergence in a 3-state environment.
Q-table initialisation	All zeros (3 × 3)	Unbiased starting point.

Q-Learning Update Rule:

$Q(s, a) = Q(s, a) + \alpha \times [\, r + \gamma \times \max Q(s', a') - Q(s, a) \,]$

At each time step the agent observes the current AQI state, selects an action via the epsilon-greedy strategy, receives the reward from the reward table, and updates the Q-value accordingly. After 500 episodes the Q-table converges and the best action for each state is extracted with argmax.

8. Training and Evaluation Results

Metric	Value
Number of states	3
Number of actions	3
Episodes used	500
Learning rate (α)	0.1
Discount factor (γ)	0.9
Exploration rate (ε)	0.1
Average reward after training	129,351.52
Final learned policy	Low→No Restriction, Medium→Partial Restriction, High→High-Pollution Alert

Learned Policy vs. Expected Policy

AQI State	Expected Action	Learned Action	Correct?
Low AQI	No Restriction	No Restriction	✓ Yes
Medium AQI	Partial Restriction	Partial Restriction	✓ Yes
High AQI	High-Pollution Alert	High-Pollution Alert	✓ Yes

All three learned actions match the expected logical actions defined by the reward table, confirming that the Q-learning agent converged to the correct policy.

9. Final Q-Table

After 500 training episodes the Q-table converged to the following values (approximate):

AQI State	No Restriction	Partial Restriction	High-Pollution Alert
Low AQI	100.0	88.0	85.0
Medium AQI	84.0	100.0	92.0
High AQI	80.0	87.0	100.0

The diagonal of the Q-table holds the highest values for each state, confirming the agent has learned the correct policy. The off-diagonal values reflect the accumulated penalties for sub-optimal actions.

10. Visualizations with Explanations

Chart 1 — Reward per Episode

This line chart plots the total reward accumulated by the agent in each of the 500 training episodes. The rewards oscillate around ~129,000–130,500, reflecting a stable but noisy environment (because of the 10 % random exploration). The overall level does not decline, confirming the agent maintains its learned policy without catastrophic forgetting. The absence of a clear upward trend is expected: the Q-table converges quickly in a 3-state problem, and subsequent variation is driven almost entirely by the random 10 % exploration.

Chart 2 — Q-Table Heatmap

The heatmap displays Q-values on a YlGnBu (yellow–green–blue) colour scale. Bright cells (≈100) lie on the diagonal, corresponding to the optimal action for each state. Darker (lower-valued) cells represent sub-optimal actions that were penalised during training. The clear visual contrast across the diagonal confirms the agent has successfully differentiated between the three AQI states and learned distinct best actions for each.

Chart 3 — Action Distribution

The bar chart shows how frequently each action appears as the best action across all three AQI states. Because there are exactly three states and each state maps to one best action, each of the three bars has a count of 1 — the agent selects every action exactly once in the final policy. This confirms that no single action dominates and that the policy is well-balanced across the AQI spectrum.

11. Answers to Assignment Questions

1. What is reinforcement learning?

Reinforcement learning (RL) is a machine-learning paradigm where an agent learns to make decisions by interacting with an environment. At each step the agent selects an action, receives a numerical reward, and updates its knowledge to maximise cumulative future reward.

2. What is the agent?

The agent is the adaptive traffic-control decision maker — the software component that reads the current AQI state and selects a traffic restriction action.

3. What is the environment?

The environment is the AQI conditions sourced from the Global Urban Air Quality Index Dataset. Each AQI record provides the current state that the agent observes.

4. Which AQI states were defined and why?

Three states were defined: Low AQI (0–100), Medium AQI (101–200), and High AQI (>200). These thresholds follow standard EPA AQI breakpoints and keep the problem tractable for an introductory tabular Q-learning implementation.

5. What are the three actions?

No Restriction (normal traffic), Partial Restriction (moderate controls), and High-Pollution Alert (strong restrictions).

6. How was the reward system designed?

The reward table was designed to reward correct matches between AQI severity and restriction level (+10) and to penalise mismatches. Over-restricting clean air (–2 to –5) and under-restricting hazardous air (–6 to –10) both incur penalties that reflect real-world cost trade-offs.

7. What is exploration vs. exploitation?

Exploration means selecting a random action (probability $\varepsilon = 0.1$) to discover potentially better strategies. Exploitation means selecting the currently known best action (probability $1-\varepsilon = 0.9$). The epsilon-greedy strategy balances both throughout training.

8. What did the final Q-table show?

The Q-table converged to ~100 on the diagonal and lower values off-diagonal, confirming the agent learned to match each AQI state with its optimal action.

9. Which action was learned for each state?

Low AQI → No Restriction, Medium AQI → Partial Restriction, High AQI → High-Pollution Alert.

10. Did the RL agent make logical decisions?

Yes. All three learned actions match the expected logical actions defined by the reward table and real-world public-health reasoning. The agent correctly learned that clean air requires no restriction and hazardous air requires the strongest alert.

11. What are the limitations?

The model uses a static next_state (no genuine state transition), only three coarse states lose granular AQI information, the reward table is hand-crafted rather than data-driven, and the environment does not simulate real traffic or pollution dynamics. More advanced approaches (Deep Q-Network, multi-step environments) would be needed for real deployment.

12. Limitations and Conclusion

Limitations:

• The next state is always set equal to the current state — no genuine state-transition dynamics are modelled.

• Only three discrete AQI states are used; fine-grained AQI variation (e.g. 101 vs 199) is lost.

• The reward table is manually designed; a data-driven reward function would be more objective.

• The simplified environment does not capture temporal dependencies or spatial variation across cities.

• Advanced RL methods (Deep Q-Network, Actor-Critic) would be needed for a production deployment.

Conclusion:

This assignment successfully demonstrated that a simple Q-learning agent can learn a logical, health-protective traffic-control policy purely from AQI data. After 500 training episodes the agent converged to the expected policy — No Restriction for clean air, Partial Restriction for moderate pollution, and High-Pollution Alert for hazardous conditions — achieving an average reward of 129,351.52. The project shows how reinforcement learning concepts (state, action, reward, episode, exploration, exploitation, Q-table) can be applied to a real-world environmental dataset.

13. Final Submission Summary

Field	Value
Student Name	MOHAMMAD QAYYUM BAIG
Registration Number	2280219
GitHub Repository	https://github.com/mohammadqbbaig/AQI-Data-Science-Project.git
Dataset Used	Global Urban AQI Dataset (2015-2025)
Number of RL States	3
Number of RL Actions	3
Episodes Used	500
Average Reward After Training	129,351.52
Best Action — Low AQI	No Restriction
Best Action — Medium AQI	Partial Restriction
Best Action — High AQI	High-Pollution Alert

14. References & AI Tool Usage Declaration

• Sutton, R.S. & Barto, A.G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press.

• US Environmental Protection Agency. AQI Basics. https://www.airnow.gov/aqi/aqi-basics/

• Global Urban Air Quality Index Dataset (2015–2025). Kaggle.

• NumPy, Pandas, and Matplotlib documentation — https://numpy.org, https://pandas.pydata.org, https://matplotlib.org

• AI Tool Usage: Claude (Anthropic) was used for guidance on report structure and explanation of RL concepts.